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SEALING DEVICE FOR BEARING UNIT

Abstract

A sealing device for sealing a bearing unit between a stationary radially outer ring and a rotary inner flanged ring against external contaminants. The sealing device includes a rotary metal first shield which has a shaped annular form and is mounted on the flanged ring, a stationary metal second shield which has a shaped annular form and is mounted on the radially outer ring and an elastomer annular sealing element, which is wholly supported by the second shield, the sealing element having an annular contacting lip and a non-contacting lip arranged radially outside the contacting lip. The non-contacting lip has a curved cross-section defined by different radii of curvature and a radially outward-facing concavity.

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Background/Summary

CROSS-REFERENCE

[0001] This application claims priority to Italian patent application no. 102024000004672 filed on Mar. 1, 2024, the contents of which are fully incorporated herein by reference.

BACKGROUND OF THE INVENTION

[0002] The present invention relates to seals, and more particularly to a sealing device for a bearing unit and the bearing unit provided with such a sealing device, preferably used in a wheel hub assembly.

[0003] Sealing devices formed as two-piece cassette seals are known, and typically include a rotary portion, for example mounted on the radially inner ring of the bearing unit, and a stationary portion, for example mounted on the radially outer ring of the bearing unit. For example, the stationary portion may include a metal shield that is interference fitted on the radially outer ring, and an elastomer portion co-molded on the metal shield. A second metal shield is interference fitted on the rotary radially inner ring. Further, the elastomer stationary portion of the seal includes one or more lips which are in contact with the rotary portion and which provide the inward seal of the bearing unit.

[0004] The typical design of a sealing device therefore includes one or more projections referred to as lips, having different shapes and orientations (typically axial and radial), which are in contact with surfaces of the rotary bearing unit. The sealing device thus prevents contaminants from entering from the outside, protecting the internal components of the bearing unit, i.e. the rolling bodies and the cages for the rolling bodies. Another important function of a sealing device is to prevent the egress of lubricants present inside the bearing unit.

[0005] A sliding friction force, and therefore energy dissipation, occurs between these contact lips and the corresponding rotary contact surfaces of the bearing unit. In general, the sealing capacity increases with the number of contact lips and the stiffness thereof. At the same time, however, as these parameters increase, so does the amount of energy dissipated.

[0006] As a result of increasingly strong global competition, there is a constant demand from customers, i.e. motor vehicle manufacturers, for continuous technical/economic improvements to wheel hub assemblies. In particular, as the technology evolves, there is increasing demand for components providing low energy-dissipation with the same performance levels as state-of-the-art solutions.

[0007] To solve this technical problem, the ideal situation would be a contactless sealing device that could still prevent the ingress of contaminants and lubricant loss, without dissipating energy. However, contactless sealing devices are often ineffective in certain circumstances, particularly in submerged environments.

[0008] It is therefore desirable to provide a sealing device with low energy-dissipation and also excellent sealing performance.

SUMMARY OF THE INVENTION

[0009] To substantially solve the technical problems set out above, one objective of the present invention is to provide a cassette sealing device including a rotary metal annular first shield and a stationary metal annular second shield, on which is mounted a sealing element provided with a contacting lip and a non-contacting lip intended to collect contaminants.

[0010] The present invention therefore provides a sealing device for a bearing unit having the features set out in the attached claims.

[0011] The present invention also relates to a bearing unit, and in particular to a bearing unit for a

wheel hub assembly provided, at the axially outer end thereof, with a sealing device according to one of the embodiments of the present invention.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0012] The invention is described below with reference to the attached drawings, which show a non-limiting example embodiment thereof, in which:

[0013] FIG. 1 is a partial cross section of a wheel hub assembly provided with a bearing unit;

[0014] FIG. 2 shows a detail, on an enlarged scale, of the wheel hub assembly in FIG. 1 including a sealing device for a bearing unit, according to an embodiment of the present invention; and

[0015] FIG. 3 shows a detail, on an even more enlarged scale, of the sealing device in FIG. 2.

DETAILED DESCRIPTION OF THE INVENTION

[0016] Purely by way of example, the present invention is described below with reference to a bearing unit **30**, preferably a motor vehicle wheel hub assembly provided with a bearing unit **30** having a sealing device **50** according to the present invention.

[0017] In FIG. 1, a bearing unit or wheel hub assembly is denoted as a whole by reference number **30**.

[0018] The bearing unit **30** has a central rotation axis X, and comprises at least a rotary radially inner flanged ring **20**, a stationary radially outer ring **31**, and a plurality of rolling bodies **32**, in this example balls, interposed or disposed between the radially outer ring **31** and the flanged ring **20**.

[0019] Throughout the present description and in the claims, terms and expressions indicating position and orientation, such as “radial” and “axial”, should be understood with reference to the central rotation axis X of the bearing unit **30**. On the other hand, expressions such as “axially outer” and “axially inner” refer to the assembled state of the wheel hub assembly, and in this case preferably refer to a wheel side and to a side opposite the wheel side, respectively.

[0020] The flanged ring **20** and the radially outer ring **31** define, between each other and at the axially outer end, a gap **35** which, if unshielded, would allow contaminants and impurities to enter the bearing unit **30**.

[0021] Therefore, in order to shield the bearing unit **30**, at least one sealing device according to the present invention is mounted inside the gap **35**. By way of example, reference is made below to a sealing device **50** mounted inside the axially outer gap **35**. With obvious variants, and without departing from the scope of the invention, a sealing device mounted inside an axially inner gap can also be defined. It should be noted that the sealing device **50** is axially symmetrical about the rotation axis X of the bearing unit **30**.

[0022] Also with reference to FIG. 2, according to an embodiment of the present invention, the sealing device **50** comprises: [0023] a rotary first shield **60** which has a shaped annular form and is preferably made of sheet metal so as to be relatively rigid, and is mounted on the flanged ring **20**; [0024] a stationary second shield **70**, which also has a shaped annular form and is preferably made of sheet metal so as to be relatively rigid, and is mounted on the radially outer ring **31**; and [0025] an annular sealing element **80**, which is wholly supported by the second shield **70**.

[0026] The first shield **60** has a mounting portion **61** interference fit on the flanged ring **20** such that, during use, is rotary about the rotation axis X. In addition, the first shield **60** includes a peripheral portion **62** radially opposite the mounting portion **61** and having a first sleeve portion **63** and a second sleeve portion **64** that each extend axially and are connected to each other by a first flange portion **65**, which extends radially. The first sleeve portion **63** is connected to the mounting portion **61** by a second flange portion **66** and is more radially proximal to the mounting portion **61** than the second sleeve portion **64**.

[0027] The second shield **70** is coaxial with the first shield **60** and has a mounting portion **71**

interference fit on the radially outer ring **31**, the second shield **70** being stationary during use. In addition, the second shield **70** includes a support portion **72** radially opposite the mounting portion **71**, the support portion **72** including a first flange portion **73**, which extends radially towards the mounting portion **61** of the first shield **60** and carries a contacting lip **81**, and a frustoconical intermediate portion **74** which connects the first flange portion **73** to the mounting portion **71** and which projects radially and axially from the mounting portion **71** towards the mounting portion **61** of the first shield **60**. The intermediate portion **74** is partially arranged radially inside the first sleeve portion **63** of the first shield **60**, the length of the intermediate portion **74** being determined by the type of application and the volume or space available for the scaling device **50**.

[0028] The mounting portion **72** is arranged radially inside the second sleeve portion **64** of the first shield **60** and is arranged at least partially inside the second sleeve portion **64**. The mounting portion **72** includes a second flange portion **75** which extends radially outwardly from the intermediate portion **74** and a sleeve portion **76** which extends axially from the second flange portion **75** from the side opposite the first shield **60** and is arranged radially adjacent to the second sleeve portion **64** of the first shield **60**. Preferably, an axially inner surface **64'** of the second sleeve portion **64** is co-planar with an axially inner surface **75'** of the second flange portion **75**.

[0029] The sealing element **80** has an annular contacting lip **81** and a non-contacting lip **90**, which is radially outside the contacting lip **81**. Both lips **81**, **90** project axially and radially from the mounting portion **72** of the second shield **70** toward the first shield **60** and are therefore inclined at an angle relative to the axis X, so as to diverge from the axis X from the side facing the first shield **60**.

[0030] In more detail and also with reference to FIG. 3, in a radial direction from the outside towards the axis X of symmetry, the sealing element **80** has and/or defines: [0031] a sleeve portion **82** which completely covers or encases the corresponding sleeve portion **76** of the second shield **70** on the side facing the first shield **60**; [0032] a deflector **83** arranged or extending radially outwardly from the sleeve portion **82** and forming or defining an axial space **10** with the second sleeve portion **64** of the first shield **60**; [0033] the axial space **10** is the only possible entry point of any contaminants and is sized relatively small or narrow in order to limit the entry of contaminants as much as possible; [0034] a radial space **11** defined between the sleeve portion **82** of the sealing element **80** and the second sleeve portion **64** of the first shield **60** along the entire length of the respective sections of the second sleeve portion **64** of the first shield **60** and of the sleeve portion **76** of the second shield **70**, the sections overlapping axially (i.e. occupying the same axial position at different radial positions); [0035] an appendage **84** which projects axially outwardly from the sleeve portion **82** of the sealing element **80**, which faces the first flange portion **65** of the first shield **60**, defining a second axial space **12**. The axial space **10**, the radial space **11** and the second axial space **12** thus form a first section of a tortuous path P which impedes the ingress of contaminants into the sealing device **50**. In addition, the appendage **84** pushes contaminants (mainly liquid) toward the first flange portion **65** of the first rotary shield **60**, to which the contaminant tends to adhere. This further impedes the path of the contaminant towards the radially inner zone of the sealing device **50**; [0036] a first chamber **15** defined between the first shield **60** and the sealing element **80** and delimited radially on the outside by the first sleeve portion **63** of the first shield **60**, and axially and radially on the inside by the non-contacting lip **90**; and [0037] the non-contacting lip **90**, provided on the sealing element **80** to interact in a non-contacting manner with the first shield **60**, has a curved cross-section defined by different radii of curvature and a radially outward-facing concavity.

[0038] The non-contacting lip **90** comprises: [0039] a rigid axially inner first portion **91** substantially completely covering the corresponding intermediate portion **74** of the second shield **70** on the side facing the first shield **60**; and [0040] a free axially outer second portion **92** connected with, and preferably integrally formed with, the first portion **91**.

[0041] The inner and outer portions **91**, **92** are delimited radially on the outside by the respective

curved surfaces **91a** and **92a**, which are joined together and have respective radii of curvature **R1** and **R2** (with $R2 < R1$). The second portion **92** faces, via a radially outer surface **92'**, the first sleeve portion **63** of the first shield **60** and, via an axially outer surface **92''**, the second flange portion **66** of the first shield **60**, defining a corresponding additional radial space **13** and a corresponding additional axial space **14** forming part of a second section of the tortuous path **P**. The second portion **92** of the non-contacting lip **90** has an angular extension α of between 80° and 100° . More precisely, the angle α is the angle between the radially outer surface **92'** of the second portion **92** of the lip **90** and a cylindrical surface **80'** of the sealing element **80** arranged radially between the contacting lip **81** and the non-contacting lip **90**.

[0042] Continuing in a radial direction from the outside towards the axis **X** of symmetry, the sealing element **80** also has and/or defines: [0043] a second chamber **16**, delimited axially on the outside by the second flange portion **66** of the first shield **60**, axially on the inside by the sealing element **80**, radially on the outside by the non-contacting lip **90** and radially on the inside by the contacting lip **81**; [0044] the contacting lip **81** which interacts slidingly with the second flange portion **66** of the first shield **60**; and [0045] a third chamber **17**, delimited radially on the outside by the contacting lip **81**, axially on the outside and radially on the inside by the mounting portion **61** of the first shield **60** and axially on the inside by a second appendage **85** which projects radially inwardly from the sealing element **80** and is used to contain the lubricating grease of the bearing unit **30**.

[0046] The sealing device **50** according to the present invention, by virtue of the features described above, enables a reduction in friction losses, not only without compromising sealing performance, but actually improving sealing performance. In fact, although on the one hand the sealing device **50** is provided with a single contacting lip **81** to minimize friction losses, on the other hand the sealing device has been designed with the aim of optimizing the efficiency of the tortuous path **P** that contaminants have to travel to enter the device, thereby drastically reducing the amount of contaminants able to reach the contacting lip **81**.

[0047] This effect is first achieved by the shape of the curved cross-section of the non-contacting lip **90**, the radially outward-facing concavity causing any contaminants entering from the first section of the tortuous path **P** (i.e., the axial space **10**, the radial space **11** and the second axial space **12**) to collect inside the first chamber **15**, and may only reach the second chamber **16** to a negligible extent (where in any case it would be contained by the contacting lip **81**). In addition, by means of adhesive effects caused by the relatively small or narrow size of the additional radial space **13** and the additional axial space **14**, any contaminants may adhere to the first shield **60** (in particular the first sleeve portion **63** and the second flange portion **66**) along a circumferential section. Finally, due to the centrifugal effect generated by the rotation of the first shield **60**, such contaminants may then leave the sealing device **50** from positions circumferentially opposite the entry positions, and in all cases via the axial space **10**. Ultimately, the shape of the non-contacting lip **90** optimizes the fluid dynamics of the seal, in particular by increasing the outward flow of contaminants.

[0048] In addition, the combination of the geometries of two components of the sealing device **50**, the rotary first shield **60** and the stationary sealing element **80**, enables the implementation of additional features which improve the overall dynamic seal: [0049] the shaped metal form of the first shield **60**, in particular the second sleeve portion **64**, has been designed to reduce the axial space **10**, which could allow contaminants to enter the radial space **11** and the second axial space **12**; [0050] the appendage **84** defined to push the contaminant towards the first shield **60**, to which contaminants tend to adhere. Such adherence to the first shield **60** further impedes the path of contaminants toward the inside of the sealing device **50** and, again due to the centrifugal forces generated during the rotation of the first shield **60**, encouraging the egress thereof from the axial space **10**; [0051] finally, especially at high speeds, the contaminant adhering to the first shield **60** exerts a force due to centrifugal and gravity effects on the non-contacting lip **90**. As a result, the

non-contacting lip **90** tends to bend, moving the axially outer surface **92** towards the respective flange portion **66** of the first shield **60**. This has a beneficial contaminant-containment effect, as the axial space **14** between the first chamber **15** and the second chamber **16** is made smaller or has a reduced volume. The elastomer used for the non-contacting lip **90**, i.e. of the scaling element **80**, must not be too flexible in order to ensure that the non-contacting lip **90** does not bend too much, thereby risking contact with the first shield **60** and the consequent increase in friction losses. The risk of such lip contact has been determined to be extremely low, since the size of the axial space **14** is preferably about 0.8 mm and the bending of the non-contacting lip **90** preferably reduces the size of the axial space **14** by only a few hundredths of a millimeter.

[0052] Ultimately, the present invention differs from the prior art in that the use of the non-contacting lip **90**, with its specific curved shape, improves the discharge of contaminants from the sealing device **50**, due to the centrifugal force applied to such contaminants by the rotation of the first shield **60**. In fact, the flow of contaminants is guided through the tortuous path P obtained by coupling the first shield **60** to the sealing element **80**, before being discharged by the centrifugal effects of the rotary first shield **60**.

[0053] Representative, non-limiting examples of the present invention were described above in detail with reference to the attached drawings. This detailed description is merely intended to teach a person of skill in the art further details for practicing preferred aspects of the present teachings and is not intended to limit the scope of the invention.

[0054] Moreover, combinations of features and steps disclosed in the above detailed description may not be necessary to practice the invention in the broadest sense, and are instead taught merely to particularly describe representative examples of the invention. Furthermore, various features of the above-described representative examples, as well as the various independent and dependent claims below, may be combined in ways that are not specifically and explicitly enumerated in order to provide additional useful embodiments of the present teachings.

[0055] All features disclosed in the description and/or the claims are intended to be disclosed separately and independently from each other for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter, independent of the compositions of the features in the embodiments and/or the claims. In addition, all value ranges or indications of groups of entities are intended to disclose every possible intermediate value or intermediate entity for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter. The invention is not restricted to the above-described embodiments, and may be varied within the scope of the following claims.

Claims

1. A sealing device for sealing from external contaminants between a stationary radially outer ring and a rotatable inner flanged ring of a bearing unit, the sealing device comprising: a first annular shield mounted on the rotatable flanged ring and formed of a metallic material; a second annular shield mounted on the stationary radially outer ring and formed of a metallic material; and an annular sealing element integrally supported by the second shield and formed of an elastomeric material, the sealing element having an annular contacting lip and a non-contacting lip located radially external to the contacting lip; wherein the non-contacting lip has a curvilinear cross-section defined by different radii of curvature and a concavity in the external radial direction.
2. The sealing device according to claim 1, wherein the non-contacting lip includes: a rigid axially inner first portion substantially completely covering a corresponding intermediate portion of the second shield on a side of the second shield facing the first annular shield; and a free axially outer second portion.
3. The sealing device according to claim 2, wherein the first portion is delimited radially outwardly by a curvilinear surface having a radius of curvature and the second portion is delimited radially

outwardly by curvilinear surface mated to the curvilinear surface of the first portion and having a radius of curvature, the radius of curvature of the curvilinear surface of the first portion being greater than the radius of curvature of the curvilinear surface of the second portion.

4. The sealing device according to claim 1, wherein the first shield has a mounting portion interference fit on the inner flanged ring and a peripheral portion radially opposite the mounting portion, the peripheral portion including a first axially-extending sleeve portion and a second axially-extending sleeve portion, the two sleeve portions being connected by a first radially extending flange portion.

5. The sealing device according to claim 4, wherein the second shield is coaxial with the first shield and has a mounting portion interference fit to the radially outer ring and a support portion radially opposite the mounting portion, the support portion including a first flange portion, the first flange portion extending radially toward the mounting portion of the first shield and carrying the contacting lip, and an intermediate truncated cone portion connecting the first flange portion to the mounting portion.

6. The sealing device according to claim 5, wherein the mounting portion of the second shield includes a second flange portion extending radially outwardly from the intermediate portion and a sleeve portion extending axially from the second flange portion.

7. The sealing device according to claim 6, wherein the sealing element has a sleeve portion completely covering the sleeve portion of the second shield on a side facing the first shield.

8. The sealing device according to claim 7, wherein the sealing element has a deflector extending radially outwardly with respect to the sleeve portion of the sealing element, an axial space being defined between the deflector and the second sleeve portion of the first shield.

9. The sealing device according to claim 7, wherein the sealing element has an appendage protruding axially outwardly from the sleeve portion of the sealing element and facing the first flange portion of the first shield.

10. A bearing unit provided with a scaling device according to claim 1, the sealing device being housed in a cavity axially external to the bearing unit, the bearing unit also including: a radially inner and rotatable flanged ring; a radially outer and stationary ring; and a plurality of rolling bodies disposed between the stationary ring and the rotatable flanged ring.
